Visualizations

2026-11-06

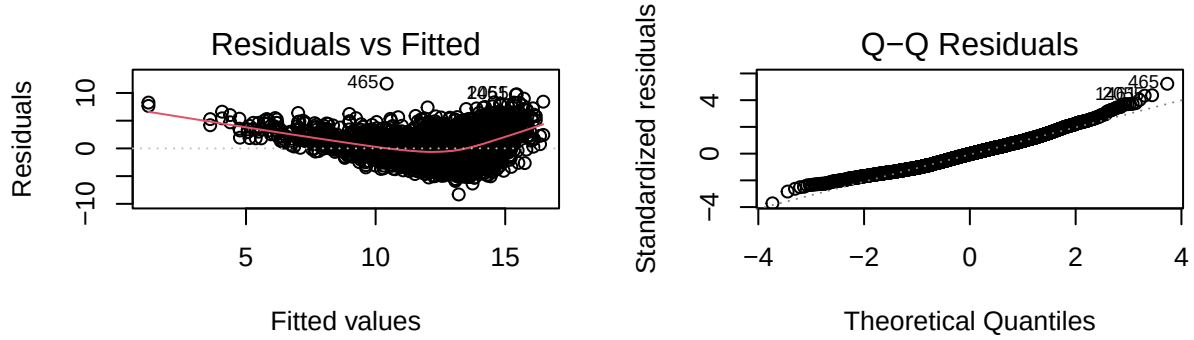
Diabetes visualizations

- #1 First we call the data and assign the names accurately in order to be able to compare.
- #2 Make the income column numeric
- #3 Test out different regression models to see what represents our data. Linear doesn't work for our model, must check other regressions
- #4 Test out quadratic regression. Looks like a match!
- #5 Lastly, create a scatter plot with a quadratic regression line to visualize the data and be able to analyze results

```
#1
new_data <- read.csv("cleaned_county_data.csv")
diabetes_data <- new_data %>%
  filter(MeasureId == "DIABETES")

#2
diabetes_data$Median_Household_Income = as.numeric(diabetes_data$Median_Household_Income)

#3
model <- lm(Data_Value ~ Median_Household_Income, data = diabetes_data)
par(mfrow = c(2, 2))
plot(model)
```

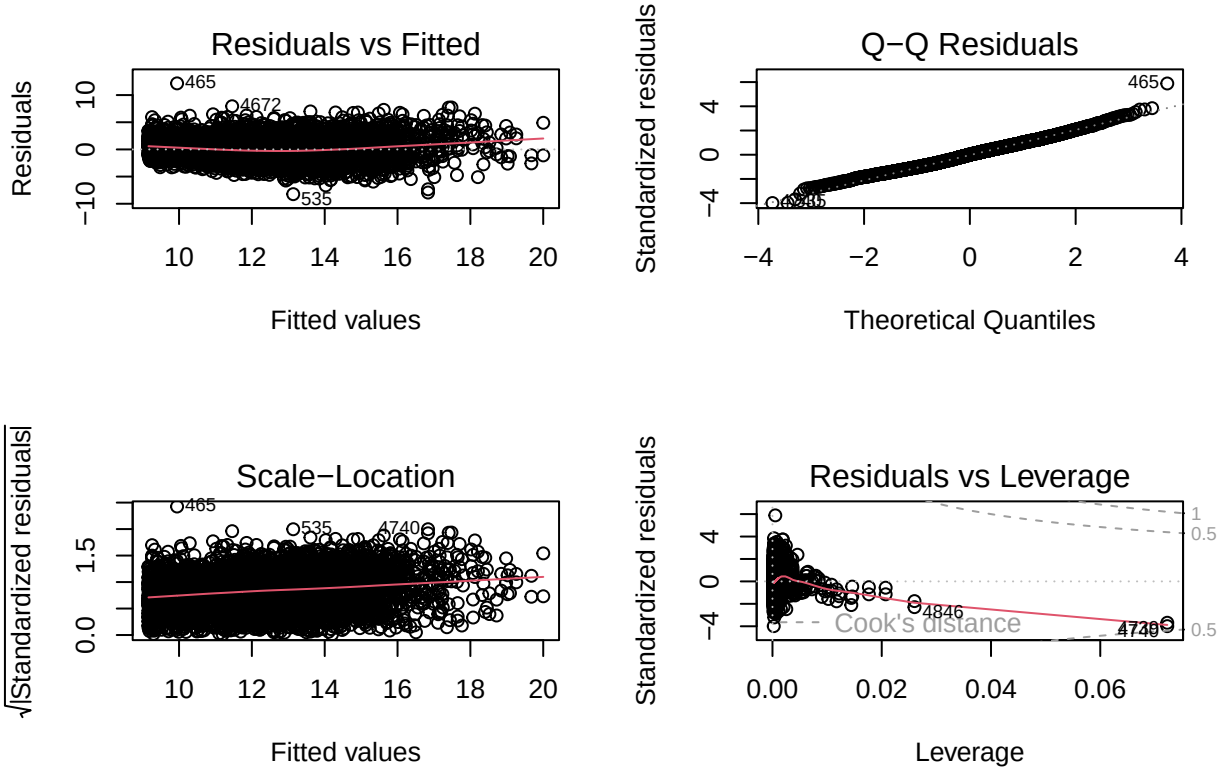



```r
#4
model2 <- lm(
  Data_Value ~ Median_Household_Income +
    I(Median_Household_Income^2),
  data = diabetes_data
  )
summary(model2)
```

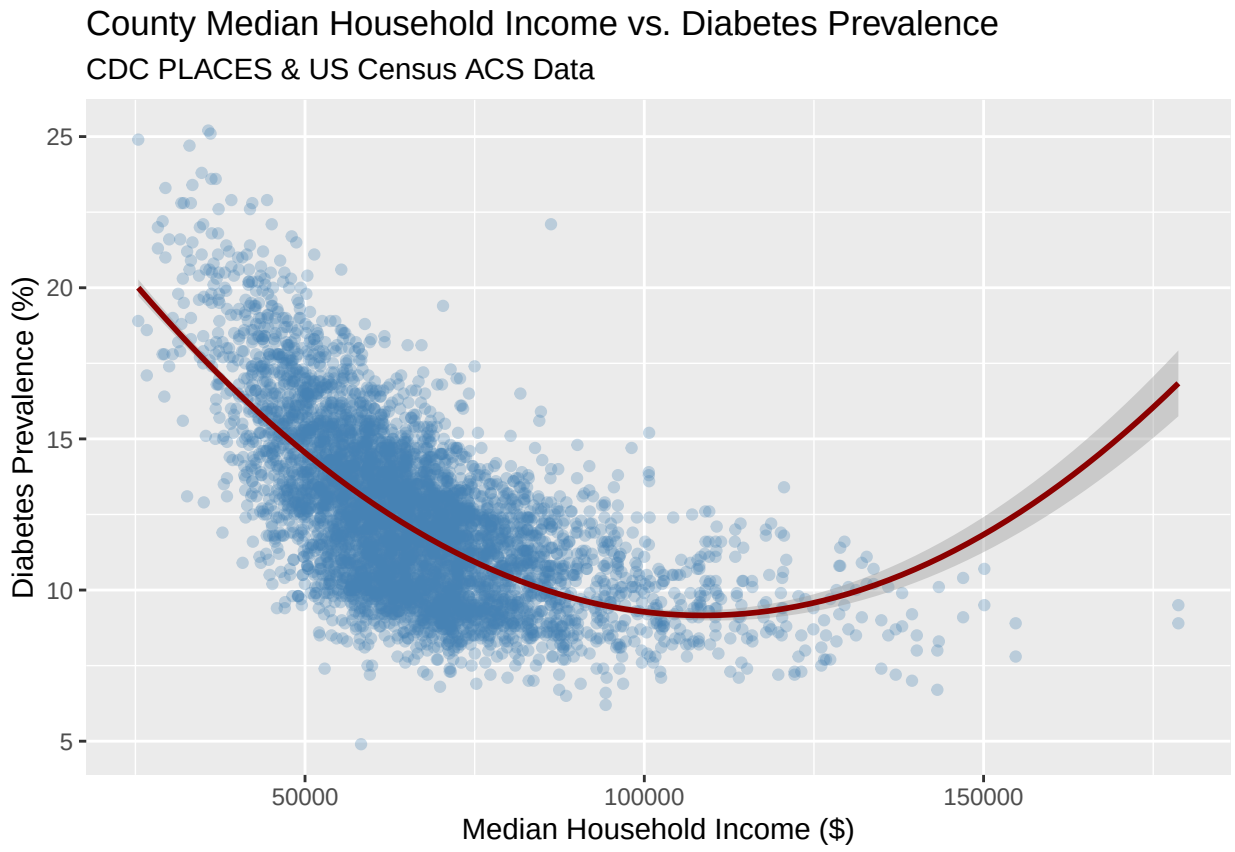
```
##
## Call:
## lm(formula = Data_Value ~ Median_Household_Income + I(Median_Household_Income^2),
##     data = diabetes_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.2355 -1.5175  0.0154  1.3843 12.1538
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   2.764e+01  3.109e-01   88.90   <2e-16 ***
## Median_Household_Income      -3.401e-04  8.207e-06  -41.43   <2e-16 ***
## I(Median_Household_Income^2)  1.565e-09  5.220e-11   29.98   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.065 on 5269 degrees of freedom
```

```
##   (6 observations deleted due to missingness)
## Multiple R-squared:  0.4497, Adjusted R-squared:  0.4495
## F-statistic:  2153 on 2 and 5269 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2))
plot(model2)
```




```r
#5
ggplot(diabetes_data, aes(x = Median_Household_Income, y = Data_Value)) +
  geom_point(alpha = 0.3, color = "steelblue") +
  geom_smooth(method = "lm", formula = y ~ x + I(x^2), color = "darkred", se = TRUE) +
  labs(
    title = "County Median Household Income vs. Diabetes Prevalence",
    subtitle = "CDC PLACES & US Census ACS Data",
    x = "Median Household Income ($)",
    y = "Diabetes Prevalence (%)"
  )
```



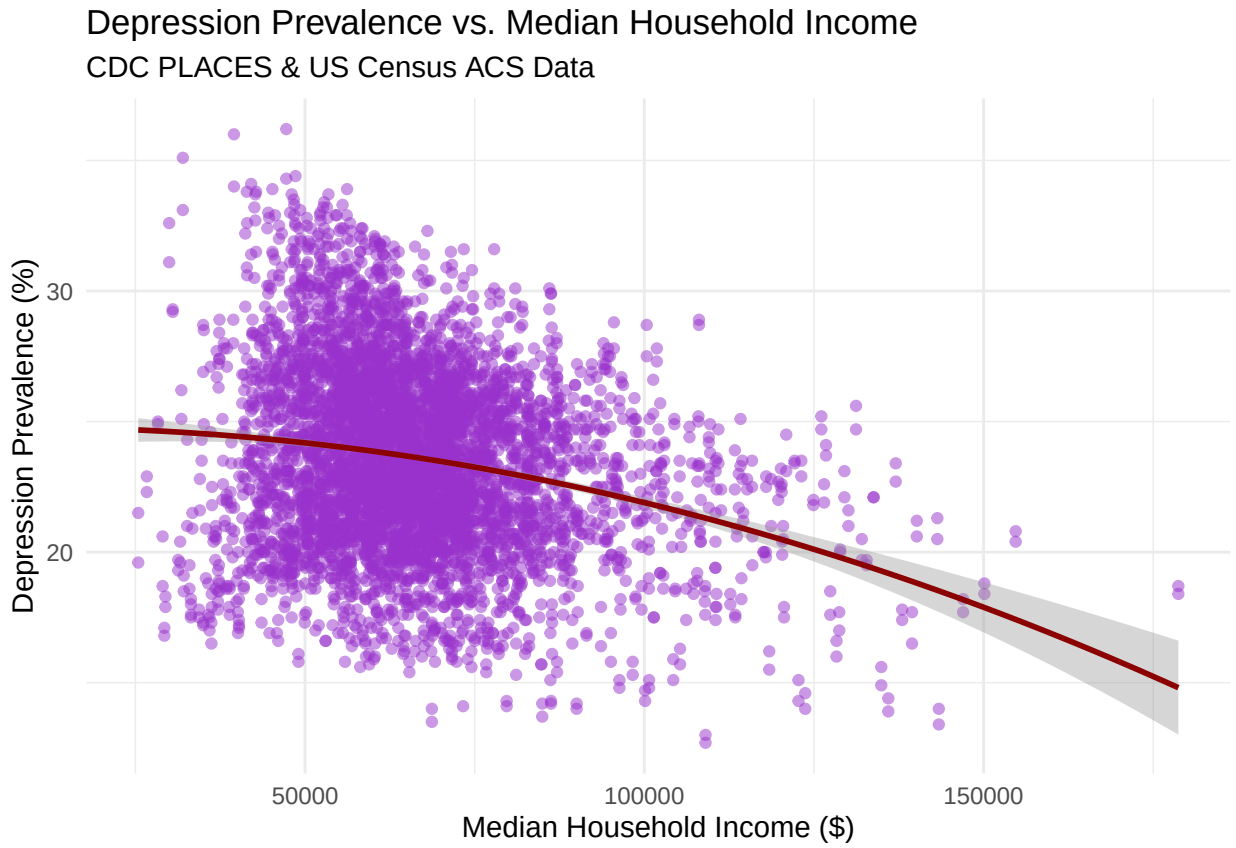
Preliminary visualizations for Depression data

- #1 Since we found our diabetes data to be quadratic, we assumed that our depression data would be quadratic aswell. We made a graph with this assumption in mind. We made a new set comparing Measure Id's of Depression in comparison to median household income. Our observations of quadratic relationship held true, no other regression models needed.

```
#1
depression_plot <- new_data %>%
  mutate(Median_Household_Income = as.numeric(as.character(Median_Household_Income))) %>%
  filter(MeasureId == "DEPRESSION") %>%
  ggplot(aes(x = Median_Household_Income, y = Data_Value)) +
  geom_point(alpha = 0.5, color = "darkorchid") +
  geom_smooth(
    method = "lm",
    formula = y ~ x + I(x^2),
    color = "darkred",
    se = TRUE
  ) +
  theme_minimal() +
  labs(
    title = "Depression Prevalence vs. Median Household Income",
    subtitle = "CDC PLACES & US Census ACS Data",
    x = "Median Household Income ($)",
    y = "Depression Prevalence (%)"
```

```
  )

print(depression_plot)
```



We have all the data cleaned, processed, and visualized. Now we can analyze our results and make sense of what they mean.